

fixed

Fixed

Revision: 43 **Last modified:** 2026-06-02T01:37:37Z **Description:** Completed workable items (with evidence references) **Authority:** Constitution §11.4.93 (workable-items DB single source of truth) **Generated-by:** scripts/docs/db_to_md.py (DB is canonical; edit via cmd/hxc-registry, not by hand)

Total completed: **222**.

HXC	Type	Pri	Title
HXC-001	Task	P1	
HXC-002	Task	P1	
HXC-003	Task	P1	
HXC-004	Task	P1	
HXC-005	Task	P1	
HXC-006	Task	P1	
HXC-007	Task	P1	
HXC-008	Task	P1	
HXC-009	Task	P1	
HXC-010	Task	P1	
HXC-011	Task	P1	
HXC-012	Task	P1	
HXC-013	Task	P1	
HXC-014	Task	P1	
HXC-015	Task	P1	
HXC-1001	Task	P0	Stand up embedded-etcd integration harness for pkg/etcd tests
HXC-1002	Feature	P0	Prove etcd Put/Get round-trip and missing-key semantics against real etcd
HXC-1003	Feature	P1	Prove etcd GetPrefix returns exactly matching multi-key set
HXC-1004	Feature	P1	Prove etcd Delete/DeletePrefix remove keys (sink-side absence)
HXC-1005	Feature	P0	Prove etcd Watch fires correct event and closes on ctx cancel
HXC-1006	Feature	P1	Prove etcd Lease/PutWithLease/RevokeLease/KeepAlive TTL behavior
HXC-1007	Feature	P0	Prove etcd distributed Lock mutual exclusion (highest-risk path)
HXC-1008	Task	P2	Add mutation-paired etcd New() default-timeout injection test
HXC-1009	Task	P2	Prove etcd error-wrapping on dead endpoint

HXC	Type	Pri	Title
HXC-1010	Feature	P0	Replace pkg/infra map-echo orchestrator with real service Boot via tests
HXC-1011	Feature	P0	Make pkg/infra Health probe real services and report false on failure
HXC-1012	Feature	P1	Make pkg/infra Logs capture real sink-side output honoring tail/since/till
HXC-1013	Feature	P1	Make pkg/infra Scale create real replicas verified via runtime query
HXC-1014	Feature	P1	Make pkg/infra VMSpawn/VMSSH establish a real session end-to-end
HXC-1015	Feature	P2	Make pkg/infra VMSimulateFailure/VMSimulatePartition assert observability
HXC-1016	Feature	P0	Add EtcdLocker integration tests against real etcd (block/acquire/lease)
HXC-1017	Bug	P0	Rewrite TestMemoryLockerConcurrent to detect a broken lock without test
HXC-1018	Task	P2	Add explicit block-then-acquire ordering test for MemoryLocker
HXC-1019	Bug	P2	Characterize or fix MemoryLocker.Lock context-cancellation goroutine
HXC-1020	Feature	P0	Wire a real Builder behind pkg/build Service and prove image is pullable
HXC-1021	Bug	P1	Drive pkg/build StateFailed via a genuine build failure not the 'fail' sent
HXC-1022	Bug	P1	Make pkg/build List/Concurrent tests Start() the service and assert termi
HXC-1023	Task	P2	Replace time.Sleep polling in pkg/build with completion sync and add s
HXC-1024	Bug	P2	Enforce or document pkg/build content-addressable cache digest integri
HXC-1025	Task	P2	Cover pkg/build cancel-mid-flight branch sets StateCancelled
HXC-1026	Feature	P0	Add real mTLS handshake test for pkg/security (valid client succeeds, n
HXC-1027	Bug	P1	Assert MinVersion==TLS13 on the client TLS path in pkg/security
HXC-1028	Feature	P0	Locate/build production Vault client and add dev-Vault integration test (
HXC-1029	Task	P3	Replace pkg/security non-nil-only TestNewTLSConfigBuilder with mea
HXC-1030	Bug	P1	Verify pkg/hxregistry item_history audit-sink rows after Create/Update
HXC-1031	Bug	P2	Assert exact and stable ComputeHeadingHash value in pkg/hxregistry
HXC-1032	Task	P2	Add pkg/hxregistry failure-path tests (dup PK, missing-id Get/Update)
HXC-1033	Bug	P2	Make pkg/hxregistry concurrent test do concurrent writes (or rename)
HXC-1034	Task	P3	Prove pkg/hxregistry timestamp round-trip and DB CHECK-constraint
HXC-1035	Bug	P0	Wire pkg/wasm WasmPlugin.Execute to real WASM output and prove tr
HXC-1036	Task	P1	Add pkg/wasm negative tests for missing execute export and corrupt mo
HXC-1037	Bug	P2	Prove pkg/wasm init/shutdown exports actually run via memory side-eff
HXC-1038	Task	P3	Cover pkg/wasm Host.Call ErrUnexpectedResult branch
HXC-1039	Bug	P1	Replace self-defeating pkg/semaphore TestSemaphore_ReleaseTooMany
HXC-1040	Feature	P1	Add pkg/semaphore M>>N concurrency invariant test under -race
HXC-1041	Task	P3	Add pkg/semaphore New(0) barrier and exact-capacity boundary tests
HXC-1042	Bug	P1	Convert pkg/storage error tests to errors.Is on exported sentinels
HXC-1043	Bug	P2	Reconcile and test pkg/storage FileStore vs MemoryStore copy semanti
HXC-1044	Task	P2	Add pkg/storage overwrite, missing-delete, and List edge tests for both s
HXC-1045	Feature	P1	Add pkg/storage -race concurrent Put/Get/Delete test to prove RWMutex
HXC-1046	Bug	P1	Gate -race in CI for pkg/metrics concurrency mutation tests
HXC-1047	Bug	P2	Complete pkg/metrics Prometheus exposition assertions (+Inf, _sum, or

HXC	Type	Pri	Title
HXC-1048	Task	P2	Add pkg/metrics duplicate-registration and histogram edge tests
HXC-1049	Task	P3	Add pkg/metrics /metrics integration test via httptest.Server
HXC-1050	Bug	P1	Test pkg/grpcutil LoggingStreamInterceptor recv/send counters
HXC-1051	Bug	P2	Strengthen pkg/grpcutil log assertions to dynamic behavior-dependent fi
HXC-1052	Task	P2	Cover pkg/grpcutil status.FromError code extraction for handler errors
HXC-1053	Bug	P1	Add pkg/context TestDetach_PreservesValues for the value-retention gu
HXC-1054	Bug	P2	Strengthen pkg/context WithTimeout deadline and expiry assertions
HXC-1055	Bug	P1	Give pkg/workerpool TestPool_NilWorkSafe_Mutation a real sink
HXC-1056	Feature	P1	Add pkg/workerpool fan-in count==N, post-Stop Submit, panic-recover, tests
HXC-1057	Bug	P1	Make pkg/errors concurrency test honest via locked GetFields path + -ra
HXC-1058	Task	P2	Pin pkg/errors Code enum exact values, distinctness, and WithFields nil
HXC-1059	Bug	P2	Replace tautological pkg/backoff cap-mutation test and pin Default()
HXC-1060	Bug	P2	Assert pkg/classads parser AST Op per operator and paren precedence
HXC-1061	Task	P3	Add pkg/classads eval failure-path and regexp non-match tests
HXC-1062	Bug	P2	Add pkg/crypto SHA-256 known-answer vectors and sentinel errors.Is c
HXC-1063	Task	P3	Add pkg/crypto AES-128/192, wrong-key decrypt, short-ciphertext, PBI
HXC-1064	Bug	P1	Add pkg/log prefix-field proof catching the SetLevel prefix-loss bug
HXC-1065	Task	P3	Strengthen pkg/log Warn test, add Debug-suppression, remove dead cap scaffolding
HXC-1066	Task	P3	Add pkg/lru boundary, cold-miss, and struct-value round-trip tests
HXC-1067	Feature	P2	Add pkg/pubsub -race concurrency test and explicit drop-boundary asse
HXC-1068	Bug	P2	Prove pkg/ratelimit bucket starts full and add Limiter conformance + -ra
HXC-1069	Bug	P1	Bring pkg/retry DoWithResult to behavioral parity with Do
HXC-1070	Task	P2	Add pkg/retry transient-recovery, jitter-bound, and computeDelay table
HXC-1071	Task	P3	Strengthen pkg/serde MustMarshal exact-content and add Unmarshal fa
HXC-1072	Bug	P2	Pin pkg/validator error contract via errors.Is and remove/wire unused se
HXC-1073	Task	P3	Add pkg/validator uint min/max, IsValidID empty, oneof/email boundar
HXC-1074	Task	P0	Establish fleet-wide -race CI gate across concurrency-sensitive packages
HXC-1075	Feature	P0	Implement NATS/JetStream client wrapper (pkg/events)
HXC-1076	Feature	P0	Implement etcd client wrapper (pkg/etcd)
HXC-1077	Feature	P0	Implement etcd discovery backend (pkg/discovery)
HXC-1078	Feature	P0	Implement real Kafka producer/consumer (pkg/pubsub)
HXC-1079	Feature	P0	Implement cgroups v2 + /proc resource reader (pkg/resources)
HXC-1080	Feature	P0	Implement Omega-model scheduler core (pkg/scheduler, internal/schedu
HXC-1081	Feature	P0	Implement optimistic-concurrency versioning for scheduler pool
HXC-1082	Feature	P1	Implement HTCondor ClassAds matching in scheduler (pkg/classads)
HXC-1083	Feature	P1	Implement gang scheduling and preemption in scheduler
HXC-1084	Feature	P0	Upgrade tmux backend to control-mode (-CC) with streaming I/O (pkg/
HXC-1085	Feature	P0	Implement CRDT session state and lifecycle (create/attach/migrate)

HXC	Type	Pri	Title
HXC-1086	Feature	P0	Implement real Node gRPC handlers serving helixv1 Node API (internal/node)
HXC-1087	Feature	P0	Make node registry etcd-backed and cross-process shared (internal/node)
HXC-1088	Feature	P0	Implement SWIM gossip membership and failure detection (pkg/swim)
HXC-1089	Feature	P0	Implement WireGuard mesh config-gen and key rotation (pkg/wireguard)
HXC-1090	Feature	P1	Implement WireGuard NAT traversal / STUN hole-punch (pkg/wireguard)
HXC-1091	Feature	P0	Wire API gateway reverse proxy with prefix routing and /health (internal/gateway)
HXC-1092	Feature	P0	Add JWT/RBAC auth middleware to API gateway (internal/gateway)
HXC-1093	Feature	P0	Implement leader election with fencing tokens (pkg/leader)
HXC-1094	Feature	P0	Implement distributed lock primitives over etcd (pkg/lock)
HXC-1095	Feature	P1	Implement Postgres-backed internal artifact registry (pkg/hxcregistry)
HXC-1096	Feature	P1	Implement GPU resource probe with /proc parser (internal/gpu)
HXC-1097	Feature	P1	Define vendor-agnostic GPUBackend interface and registry (internal/gpu)
HXC-1098	Feature	P2	Implement GPU sharing modes (MPS, time-slice, MIG, exclusive)
HXC-1099	Feature	P1	Implement Build Service orchestrator and worker pool (internal/build)
HXC-1101	Feature	P0	Implement e2ee transport record protocol (security/pkg/e2ee)
HXC-1102	Feature	P1	Implement software-rooted attestation (security/pkg/attestation)
HXC-1103	Feature	P0	Implement RBAC scopes and identity bindings (internal/security)
HXC-1104	Feature	P1	Implement SPIFFE/SPIRE SVID issuance and mTLS identity (internal/security)
HXC-1105	Feature	P0	Implement htmux CLI raw-mode terminal client (cmd/htmux)
HXC-1106	Feature	P0	Implement session I/O PTY-over-WebSocket forwarding (cmd/helix-session)
HXC-1107	Feature	P0	Implement WebSocket envelope and MessagePack framing (pkg/websocket)
HXC-1108	Feature	P1	Implement HelixQA live-service Challenge runner (cmd/helix-test)
HXC-1109	Feature	P1	Implement deterministic-sim, chaos, and device-sim test harness (cmd/helix-test)
HXC-1110	Task	P1	Wire Prometheus /metrics endpoint uniformly across all services (cmd/helix-test)
HXC-1111	Task	P1	Wire Jaeger/OTel trace-ID propagation across services (pkg/tracing)
HXC-1112	Task	P1	Wire pkg/health gRPC health protocol on every service (pkg/health, cmd/helix-test)
HXC-1114	Feature	P0	Provision PostgreSQL primary schema with 15 tables, indexes, and triggers
HXC-1115	Task	P0	Define etcd key namespace and key-builder constants (pkg/etcd)
HXC-1116	Feature	P1	Define Redis cache, routing, and rate-limit key structure (pkg/storage)
HXC-1118	Feature	P1	Implement REST API surface on API Gateway with OpenAPI 3.0
HXC-1120	Task	P1	Configure Kafka topics with partitions/replication/retention
HXC-1121	Task	P0	Configure NATS JetStream streams (HELIX_NODES/SESSIONS/SCHEDULE/HEALTH/ALERTS)
HXC-1122	Feature	P2	Implement Cap'n Proto zero-copy serialization for control-plane (pkg/serialization)
HXC-1123	Feature	P2	Implement FlatBuffers serialization for GPU compute payloads
HXC-1124	Feature	P2	Implement ZeroMQ message patterns for internal data planes
HXC-1125	Feature	P1	Implement Setup Wizard single-command node onboarding (cmd, BASIS)
HXC-1127	Feature	P1	Implement OPA/WASM Policy Engine with HelixConstitution enforcement
HXC-1130	Feature	P2	Implement Zellij and GNU screen session backends
HXC-1131	Feature	P2	Implement Backup Service (etcd snapshots, WAL archival, Ceph checkpoints)

HXC	Type	Pri	Title
HXC-1132	Feature	P1	Implement Metrics Collector node scrape endpoint and GPU metrics aggregation
HXC-1133	Feature	P1	Implement token-bucket rate limiter for API gateway (pkg/ratelimit)
HXC-1134	Feature	P2	Implement Vault secret injection and rotation for all services
HXC-1135	Feature	P0	Implement two-node cluster formation end-to-end (exit gate)
HXC-1136	Feature	P0	Implement session-create end-to-end path (htmux new exit gate)
HXC-1137	Task	P1	Benchmark job scheduling decision latency under 100ms (exit gate)
HXC-1138	Task	P1	Enforce >60% pkg/ line coverage and paired mutation tests (Constitution)
HXC-1141	Feature	P2	Implement interactive-mode AI CLI agent resource provisioning
HXC-1142	Feature	P2	Implement hybrid mode switching and batch/interactive resource sharing
HXC-1144	Feature	P2	Implement WireGuard mesh network policies (segmentation)
HXC-1148	Feature	P1	Replace pkg/wireguard NAT discovery with RFC 5389 STUN client
HXC-1149	Task	P3	Return typed ErrUnsupported for UPnP/NAT-PMP in pkg/wireguard
HXC-1150	Feature	P1	Add WireGuard key-rotation grace/overlap window with ActiveKeys()
HXC-1151	Task	P1	Add pkg/discovery etcd-backed integration test against a real etcd
HXC-1152	Feature	P1	Implement pkg/session container-checkpoint migration strategy (CRDT)
HXC-1156	Feature	P1	Implement thermal-aware scheduler plugin (throttle on overheat)
HXC-1157	Feature	P1	Implement semi-trusted node output verification (redundant-compute/chaos)
HXC-1160	Feature	P2	Improve tmux session backend to full PTY wrap on Attach
HXC-1162	Task	P2	E2E: multi-node session CRDT convergence
HXC-1188	Feature	P0	Enforce work-unit resource limits (duration/memory/CPU) on edge devices
HXC-1189	Feature	P0	Implement EdgeAwarePlugin scheduler Filter stage
HXC-1190	Feature	P1	Implement EdgeAwarePlugin scheduler Score stage
HXC-1191	Feature	P1	Implement declarative per-tier ScheduleRule engine
HXC-1195	Feature	P0	Implement edge trust-level model (STANDARD/SEMI/EDGE_DONOR)
HXC-1196	Feature	P0	Enforce workload restriction matrix by trust level
HXC-1198	Feature	P1	Implement offline sync protocol with delta compression
HXC-1199	Feature	P1	Implement edge sensor-fusion framework and stream workload type
HXC-1210	Bug	P0	Wire EtcdBackend into the node agent for cluster-wide discovery
HXC-1211	Bug	P0	Make internal/scheduler.StreamJobEvents a real event stream
HXC-1212	Feature	P0	Connect pkg/classads to scheduler Filter/Score stages
HXC-1213	Feature	P0	Add auth + RBAC enforcement to the gateway
HXC-1214	Feature	P1	Add S3-compatible (minio) backend to pkg/storage
HXC-1215	Feature	P1	Add OTLP/Jaeger export shim for pkg/tracing
HXC-1220	Task	P2	Replace custom policy engine with OPA/Rego
HXC-1229	Task	P0	Build device profile registry with versioned YAML schema for T1-T8
HXC-1231	Feature	P1	Implement device provisioning lifecycle abstraction (Provisioner/Instantiator)
HXC-1236	Feature	P0	Implement INetwork/HelixNetwork trait with production and simulation
HXC-1237	Feature	P0	Wire DST sim transport seam into pkg/swim membership (prod+sim parallel)
HXC-1238	Feature	P1	Implement BUGGIFY macro framework with ~25% deterministic fire rate

HXC	Type	Pri	Title
HXC-1239	Feature	P1	Implement 10:1 DST virtual-time compression by fast-forwarding idle p
HXC-1240	Feature	P1	Achieve 1,000+ simulated nodes in a single DST process
HXC-1241	Feature	P1	Author DST consensus+gossip workloads using SETUP→EXECUTION→CHECK→METRICS pattern
HXC-1243	Feature	P0	Implement 8 network fault injectors (latency, loss, corruption, reorder, d partition, DNS, TCP reset)
HXC-1244	Feature	P0	Implement 8 node fault injectors (VM crash/restart/pause, CPU/mem/di kill, graceful shutdown)
HXC-1247	Feature	P1	Implement pure-Go in-sim chaos faults toward 25+ (ClockSkew, DiskFi MessageReorder, Byzantine, etc.)
HXC-1248	Feature	P0	Build YAML chaos Scenario Engine with phases, blast radius and abort-
HXC-1249	Feature	P0	Implement emergency-stop and auto-recovery with <=2s halt latency
HXC-1254	Feature	P1	Implement TestRunner with parallel suite execution and result collection
HXC-1255	Feature	P1	Implement session test state machine (IDLE→SETUP→RUNNING→CHAOS_INJECT→VERIFY→RECO
HXC-1260	Feature	P1	Implement metrics validation against baseline KPI table with severity ga
HXC-1261	Feature	P1	Implement Welch's t-test statistical regression detector for HelixQA
HXC-1265	Feature	P1	Implement capability-based plugin sandbox with resource limits and auc
HXC-1267	Feature	P0	Implement cmd/helix-test CLI with dst/chaos/device/snapshot subcomm
HXC-1268	Feature	P1	Build cmd/helix-snapshot standalone CLI reusing pkg/testing/snapshot
HXC-1278	Research	P3	Add property-based + coverage-guided fuzzing for cluster operations
HXC-1290	Feature	P0	Implement pkg/device universal capability Descriptor with 15-tier + trus
HXC-1291	Feature	P0	Implement device.Probe(ctx) arch/CPU/mem auto-detection with overri
HXC-1292	Feature	P0	Implement device.AssignTier decision tree per §50 taxonomy (riscv64-> fw->T1, GPU+battery->T9)
HXC-1296	Feature	P0	Implement device.Descriptor <-> pb.Node round-trip mapping helpers
HXC-1297	Feature	P0	Implement pkg/scheduler tier-aware + power-aware Plugin (tier_power.
HXC-1298	Feature	P0	Implement power-aware T9 HANDHELD scheduling policy (battery/th active gating)
HXC-1299	Feature	P0	Implement pkg/discovery 15-tier registry + SelectByTier(tier, minTrust)
HXC-1301	Feature	P1	Implement pkg/cloudspot PreemptionWatcher with injectable SignalSou hooks
HXC-1302	Feature	P1	Extend pkg/wireguard with preemption-safe teardown / spot-drain hook
HXC-1303	Feature	P1	Implement pkg/inference provider-agnostic Backend interface + capabil
HXC-1304	Feature	P1	Implement pkg/inference LocalBackend wrapping internal/llm manager
HXC-1310	Feature	P0	Implement per-tier security model enforcement (sandbox + data/network level)
HXC-1311	Docs	P1	Author complete YAML tier definitions (T1-T15) with min_requirement
HXC-1334	Feature	P0	Implement pkg/hlc Hybrid Logical Clock with drift clamp and total orde
HXC-1335	Feature	P0	Implement pkg/crdt G-Counter with per-replica max-merge convergence
HXC-1336	Feature	P0	Implement pkg/crdt LWW-Register with HLC last-writer-wins tiebreak
HXC-1337	Feature	P0	Implement pkg/crdt OR-Set with unique-tag add/remove semantics

HXC	Type	Pri	Title
HXC-1338	Feature	P1	Implement pkg/crdt/merkle anti-entropy tree with divergent-key Diff
HXC-1339	Feature	P1	Implement Vector Clock causality type for Tier-2 operational state
HXC-1340	Feature	P0	Implement pkg/federation Cell model and lifecycle state machine
HXC-1341	Feature	P0	Implement thread-safe Cell Registry with add/remove/list/snapshot
HXC-1342	Feature	P1	Implement pkg/federation federated API aggregation with partial failure
HXC-1343	Feature	P1	Implement Phi-accrual failure detector in pkg/swim
HXC-1353	Feature	P1	Implement WireGuard zero-downtime key rotation with overlap window
HXC-1365	Feature	P1	Extend pkg/scheduler with latency-aware spot/preemptible scoring
HXC-1373	Feature	P1	Implement split-brain prevention and partition-classification logic
HXC-902	Task	P1	
HXC-903	Task	P1	